SG No. 148 C_{3i}^{2} $R\bar{3}$ [trigonal]

* plus set: $+[0,\,0,\,0]$, $+\left[\frac{2}{3},\,\frac{1}{3},\,\frac{1}{3}\right]$, $+\left[\frac{1}{3},\,\frac{2}{3},\,\frac{2}{3}\right]$
* Wyckoff site: `3a`, site symmetry: `-3.`

Table 1: Wyckoff bond: `3a@3a`, bond symmetry: `-3.`

No.	vector	center	mapping
1	$[0,\,0,\,Z]$	$[0,\,0,\,0]$	`[1,2,3,-4,-5,-6]`

Table 2: Wyckoff bond: `9b@3a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[0,\,0,\,0]$	`[1,-4]`
2	$[-Y,\,X-Y,\,Z]$	$[0,\,0,\,0]$	`[2,-5]`
3	$[-X+Y,\,-X,\,Z]$	$[0,\,0,\,0]$	`[3,-6]`

* Wyckoff site: `3b`, site symmetry: `-3.`

Table 3: Wyckoff bond: `3a@3b`, bond symmetry: `-3.`

No.	vector	center	mapping
1	$[0,\,0,\,Z]$	$\left[0,\,0,\,\frac{1}{2}\right]$	`[1,2,3,-4,-5,-6]`

Table 4: Wyckoff bond: `9b@3b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[0,\,0,\,\frac{1}{2}\right]$	`[1,-4]`
2	$[-Y,\,X-Y,\,Z]$	$\left[0,\,0,\,\frac{1}{2}\right]$	`[2,-5]`
3	$[-X+Y,\,-X,\,Z]$	$\left[0,\,0,\,\frac{1}{2}\right]$	`[3,-6]`

* Wyckoff site: `6c`, site symmetry: `3.`

Table 5: Wyckoff bond: `6a@6c`, bond symmetry: `3.`

No.	vector	center	mapping
1	$[0,\,0,\,Z]$	$[0,\,0,\,z]$	`[1,2,3]`
2	$[0,\,0,\,-Z]$	$[0,\,0,\,-z]$	`[4,5,6]`

Table 6: Wyckoff bond: `18b@6c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[0,\,0,\,z]$	`[1]`
2	$[-Y,\,X-Y,\,Z]$	$[0,\,0,\,z]$	`[2]`
3	$[-X+Y,\,-X,\,Z]$	$[0,\,0,\,z]$	`[3]`
4	$[-X,\,-Y,\,-Z]$	$[0,\,0,\,-z]$	`[4]`
5	$[Y,\,-X+Y,\,-Z]$	$[0,\,0,\,-z]$	`[5]`

continued ...

Table 6

No.	vector	center	mapping
6	$[X-Y,\ X,\ -Z]$	$[0,\ 0,\ -z]$	`[6]`

* Wyckoff site: `9d`, site symmetry: `-1`

Table 7: Wyckoff bond: `9a@9d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,-6]`

* Wyckoff site: `9e`, site symmetry: `-1`

Table 8: Wyckoff bond: `9a@9e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,-6]`

* Wyckoff site: `18f`, site symmetry: `1`

Table 9: Wyckoff bond: `18a@18f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[-y,\ x-y,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[-x+y,\ -x,\ z]$	`[3]`
4	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	`[4]`
5	$[Y,\ -X+Y,\ -Z]$	$[y,\ -x+y,\ -z]$	`[5]`
6	$[X-Y,\ X,\ -Z]$	$[x-y,\ x,\ -z]$	`[6]`